I- Logarithms in different bases

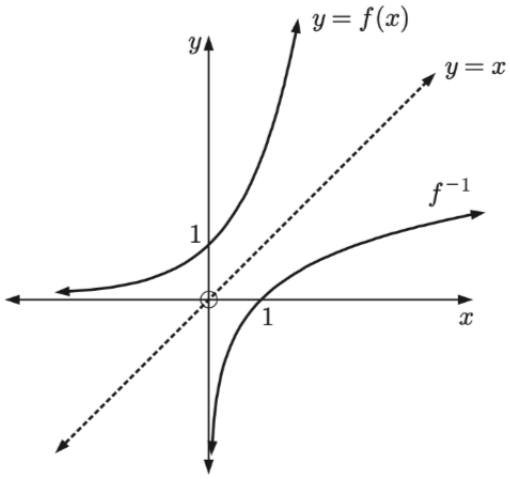
a) Logarithms in base 10

Consider the exponential function $f : x \mapsto 10^x$
or $f(x) = 10^x$.

The graph of $y = f(x)$ is shown alongside, along with its inverse function f^{-1}.

Since f is defined by $y = 10^x$,
f^{-1} is defined by $x = 10^y$.
{interchanging x and y}

y is the exponent to which the base 10 is raised in order to get x.



Definition

The logarithm in base 10 of a positive number is the exponent when the number is written as a power of 10.

Property

If $f(x) = 10^x$, then $f^{-1}(x) = \log_{10} x$.

Remark. If no base is indicated, we assume it means base 10. $\log a$ means $\log_{10} a$.

Property

$\log 10^x = x$ for any $x \in \mathbb{R}$.

Example

Without using a calculator, find :

1. $\log 100$
2. $\log\left(\sqrt[4]{10}\right)$

EXERCISE 1

Without using a calculator, find:

a $\log 10\,000$ **b** $\log 0.001$ **c** $\log 10$ **d** $\log 1$

e $\log \sqrt{10}$ **f** $\log(\sqrt[3]{10})$ **g** $\log\left(\frac{1}{\sqrt[4]{10}}\right)$ **h** $\log\left(10\sqrt{10}\right)$

i $\log \sqrt[3]{100}$ **j** $\log\left(\frac{100}{\sqrt{10}}\right)$ **k** $\log\left(10 \times \sqrt[3]{10}\right)$ **l** $\log\left(1000\sqrt{10}\right)$

EXERCISE 2

Simplify:

a $\log 10^n$ **b** $\log\left(10^a \times 100\right)$ **c** $\log\left(\frac{10}{10^m}\right)$ **d** $\log\left(\frac{10^a}{10^b}\right)$

Property

$a = 10^{\log a}$ for any $a > 0$.

Example

Use your calculator to write the following in the form 10^x where x is correct to 4 decimal places :

1. 8
2. 800
3. 0.08

EXERCISE 3

Use your calculator to write the following in the form 10^x where x is correct to 4 decimal places:

a 6 **b** 60 **c** 6000 **d** 0.6 **e** 0.006

f 15 **g** 1500 **h** 1.5 **i** 0.15 **j** 0.000 15

Example

1. Use your calculator to find :
   (a) $\log 2$
   (b) $\log 20$
2. Explain why $\log 20 = \log 2 + 1$.

EXERCISE 4

a Use your calculator to find: **i** $\log 5$ **ii** $\log 0.05$
b Explain why $\log 0.05 = \log 5 - 2$.

Example

Find x if :

1. $\log x = 3$
2. $\log x \approx -0.271$

EXERCISE 5

Find x if:

a $\log x = 2$ **b** $\log x = 1$ **c** $\log x = 0$
d $\log x = -1$ **e** $\log x = \frac{1}{2}$ **f** $\log x = -\frac{1}{2}$
g $\log x = 4$ **h** $\log x = -5$ **i** $\log x \approx 0.8351$
j $\log x \approx 2.1457$ **k** $\log x \approx -1.378$ **l** $\log x \approx -3.1997$

b) Logarithms in base a

Definition

If $b = a^x$, $a \neq 1$, $a > 0$, we say that x is the logarithm in base a of b, and that

$$b = a^x \iff x = \log_a b, \quad b > 0.$$

Property

f $f(x) = a^x$ then $f^{-1}(x) = \log_a x$.

Property

$$x = \log_a(a^x).$$

and

$$x = a^{\log_a x} \qquad \text{provided } x > 0.$$

Proof

If $y = a^x$ then $x = \log_a y$

If $x = a^y$ then $y = \log_a x$

Example

1. Write an equivalent exponential equation for $\log_{10} 1000 = 3$.
2. Write an equivalent logarithmic equation for $3^4 = 81$.

EXERCISE 6

Write an equivalent exponential equation for:

a $\log_{10} 100 = 2$ b $\log_{10} 10\,000 = 4$ c $\log_{10}(0.1) = -1$

d $\log_{10} \sqrt{10} = \frac{1}{2}$ e $\log_2 8 = 3$ f $\log_3 9 = 2$

g $\log_2(\frac{1}{4}) = -2$ h $\log_3 \sqrt{27} = 1.5$ i $\log_5(\frac{1}{\sqrt{5}}) = -\frac{1}{2}$

EXERCISE 7

Write an equivalent logarithmic equation for:

a $2^2 = 4$	b $4^3 = 64$	c $5^2 = 25$
d $7^2 = 49$	e $2^6 = 64$	f $2^{-3} = \frac{1}{8}$
g $10^{-2} = 0.01$	h $2^{-1} = \frac{1}{2}$	i $3^{-3} = \frac{1}{27}$

Example

1. $\log_2 16$
2. $\log_5 0.2$
3. $\log_{10}\left(\sqrt[5]{100}\right)$
4. $\log_2\left(\dfrac{1}{\sqrt{2}}\right)$

EXERCISE 8

Find:

a $\log_{10} 100\,000$	b $\log_{10}(0.01)$	c $\log_3 \sqrt{3}$	d $\log_2 8$
e $\log_2 64$	f $\log_2 128$	g $\log_5 25$	h $\log_5 125$
i $\log_2(0.125)$	j $\log_9 3$	k $\log_4 16$	l $\log_{36} 6$
m $\log_3 243$	n $\log_2 \sqrt[3]{2}$	o $\log_a a^n$	p $\log_8 2$
q $\log_t\left(\frac{1}{t}\right)$	r $\log_6 6\sqrt{6}$	s $\log_4 1$	t $\log_9 9$

EXERCISE 9

Solve for x:

a $\log_2 x = 3$	b $\log_4 x = \frac{1}{2}$	c $\log_x 81 = 4$	d $\log_2(x-6) = 3$

EXERCISE 10

Simplify:

a $\log_4 16$	b $\log_2 4$	c $\log_3\left(\frac{1}{3}\right)$	d $\log_{10}\sqrt[4]{1000}$
e $\log_7\left(\frac{1}{\sqrt{7}}\right)$	f $\log_5(25\sqrt{5})$	g $\log_3\left(\frac{1}{\sqrt{27}}\right)$	h $\log_4\left(\frac{1}{2\sqrt{2}}\right)$
i $\log_x x^2$	j $\log_x \sqrt{x}$	k $\log_m m^3$	l $\log_x(x\sqrt{x})$
m $\log_n\left(\frac{1}{n}\right)$	n $\log_a\left(\frac{1}{a^2}\right)$	o $\log_a\left(\frac{1}{\sqrt{a}}\right)$	p $\log_m \sqrt{m^5}$

II- Properties of logarithms

a) Laws of logarithms

Investigation

1. Use your calculator to find :
   (a) $\log 2 + \log 3$
   (b) $\log 3 + \log 7$
   (c) $\log 4 + \log 20$
   (d) $\log 6$
   (e) $\log 21$
   (f) $\log 80$

   From your answers, suggest a possible simplification for $\log a + \log b$.
2. Use your calculator to find :
   (a) $\log 6 - \log 2$
   (b) $\log 12 - \log 3$
   (c) $\log 3 - \log 5$
   (d) $\log 3$
   (e) $\log 4$
   (f) $\log(0.6)$

   From your answers, suggest a possible simplification for $\log a - \log b$.
3. Use your calculator to find :
   (a) $3 \log 2$
   (b) $2 \log 5$
   (c) $-4 \log 3$
   (d) $\log(2^3)$
   (e) $\log(5^2)$
   (f) $\log(3^{-4})$

   From your answers, suggest a possible simplification for $n \log a$.

Property

In any base c where $c \neq 1$ and $c > 0$, if A and B are both positive, then :

— $\log_c A + \log_c B = \log_c(AB)$

— $\log_c A - \log_c B = \log_c\left(\frac{A}{B}\right)$

— $n \log_c A = \log_c(A^n)$

Proof

—

$$\begin{aligned} c^{\log_c A + \log_c B} &= c^{\log_c A}\, c^{\log_c B} \\ &= A \times B \\ &= c^{\log_c(AB)}. \end{aligned}$$

Hence,

$$\log_c A + \log_c B = \log_c(AB).$$

—

$$\begin{aligned} c^{\log_c A - \log_c B} &= \frac{c^{\log_c A}}{c^{\log_c B}} \\ &= \frac{A}{B} \\ &= c^{\log_c\left(\frac{A}{B}\right)}. \end{aligned}$$

Hence,

$$\log_c A - \log_c B = \log_c\left(\frac{A}{B}\right).$$

—

$$\begin{aligned} c^{n \log_c A} &= \left(c^{\log_c A}\right)^n \\ &= A^n \\ &= c^{\log_c(A^n)}. \end{aligned}$$

Hence,

$$n \log_c A = \log_c(A^n).$$

Example

Use the laws of logarithms to write the following as a single logarithm or as an integer :

1. $\log 5 + \log 3$
2. $\log_3 24 - \log_3 8$
3. $\log_2 5 - 1$

Example

Simplify by writing as a single logarithm or as a rational number :

1. $2\log 7 - 3\log 2$
2. $2\log 3 + 3$
3. $\dfrac{\log 8}{\log 4}$

EXERCISE 11

Write as a single logarithm or integer:

a $5\log 2 + \log 3$ **b** $2\log 3 + 3\log 2$ **c** $3\log 4 - \log 8$

d $2\log_3 5 - 3\log_3 2$ **e** $\frac{1}{2}\log_6 4 + \log_6 3$ **f** $\frac{1}{3}\log\left(\frac{1}{8}\right)$

g $3 - \log 2 - 2\log 5$ **h** $1 - 3\log 2 + \log 20$ **i** $2 - \frac{1}{2}\log_n 4 - \log_n 5$

EXERCISE 12

Simplify without using a calculator:

a $\dfrac{\log 4}{\log 2}$ **b** $\dfrac{\log_5 27}{\log_5 9}$ **c** $\dfrac{\log 8}{\log 2}$ **d** $\dfrac{\log 3}{\log 9}$ **e** $\dfrac{\log_3 25}{\log_3(0.2)}$ **f** $\dfrac{\log_4 8}{\log_4(0.25)}$

Example

Show that :

1. $\log\left(\frac{1}{9}\right) = -2\log 3$
2. $\log 500 = 3 - \log 2$

EXERCISE 13

Show that:

a $\log 9 = 2\log 3$ **b** $\log\sqrt{2} = \frac{1}{2}\log 2$ **c** $\log\left(\frac{1}{8}\right) = -3\log 2$

d $\log\left(\frac{1}{5}\right) = -\log 5$ **e** $\log 5 = 1 - \log 2$ **f** $\log 5000 = 4 - \log 2$

EXERCISE 14

If $\log_2 P = x$, $\log_2 Q = y$, and $\log_2 R = z$ write in terms of x, y, and z:

a $\log_2(PR)$ **b** $\log_2(RQ^2)$ **c** $\log_2\left(\frac{PR}{Q}\right)$

d $\log_2(P^2\sqrt{Q})$ **e** $\log_2\left(\frac{Q^3}{\sqrt{R}}\right)$ **f** $\log_2\left(\frac{R^2\sqrt{Q}}{P^3}\right)$

b) Equations of logarithms

Example

Write these as logarithmic equations (in base 10) :

1. $y = a^2 b$
2. $y = \dfrac{a}{b^3}$
3. $P = \dfrac{20}{\sqrt{n}}$

Example

Write the following equations without logarithms :

1. $\log A = \log b + 2\log c$
2. $\log_2 M = 3\log_2 a - 2$

EXERCISE 15

a Write $y = 3 \times 2^x$ as a logarithmic equation in base 2.
b Hence write x in terms of y.
c Find x when:
i $y = 3$ ii $y = 12$ iii $y = 30$

EXERCISE 16

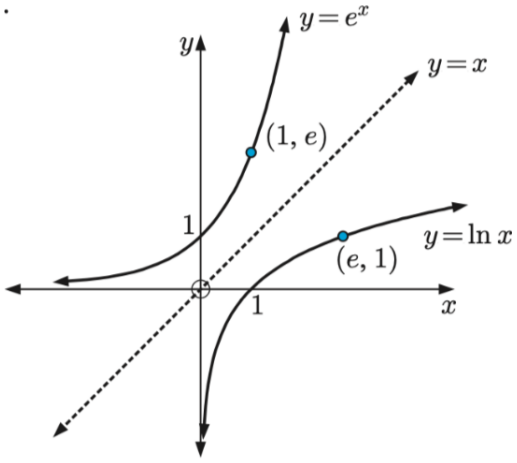
Solve for x:

a $\log_3 27 + \log_3(\frac{1}{3}) = \log_3 x$
b $\log_5 x = \log_5 8 - \log_5(6 - x)$
c $\log_5 125 - \log_5 \sqrt{5} = \log_5 x$
d $\log_{20} x = 1 + \log_{20} 10$
e $\log x + \log(x + 1) = \log 30$
f $\log(x + 2) - \log(x - 2) = \log 5$

III- Natural logarithms

Given the exponential function $f(x) = e^x$, the inverse function $f^{-1} = \log_e x$ is the logarithm in base e.

We use $\ln x$ to represent $\log_e x$, and call $\ln x$ the **natural logarithm** of x.



Definition

$\ln(e^x) = x$ and $e^{\ln x} = x$.

Property

$a^x = e^{x \ln a}$, $a > 0$.

Proof

$a^x = (e^{\ln a})^x = e^{x \ln a}$,

Example

Use your calculator to write the following in the form e^k where k is correct to 4 decimal places :

1. 50
2. 0.005

EXERCISE 17

Without using a calculator find:

a $\ln e^2$ **b** $\ln e^3$ **c** $\ln \sqrt{e}$ **d** $\ln 1$

e $\ln\left(\frac{1}{e}\right)$ **f** $\ln \sqrt[3]{e}$ **g** $\ln\left(\frac{1}{e^2}\right)$ **h** $\ln\left(\frac{1}{\sqrt{e}}\right)$

EXERCISE 18

Explain why $\ln(-2)$ and $\ln 0$ cannot be found.

EXERCISE 19

Simplify:

a $\ln e^a$ **b** $\ln(e \times e^a)$ **c** $\ln\left(e^a \times e^b\right)$ **d** $\ln(e^a)^b$ **e** $\ln\left(\frac{e^a}{e^b}\right)$

Example

Find x if :

1. $\ln x = 2.17$
2. $\ln x = -0.384$

EXERCISE 20

Find x if:

a $\ln x = 3$ **b** $\ln x = 1$ **c** $\ln x = 0$ **d** $\ln x = -1$

e $\ln x = -5$ **f** $\ln x \approx 0.835$ **g** $\ln x \approx 2.145$ **h** $\ln x \approx -3.2971$

Example

Use the laws of logarithms to write the following as a single logarithm :

1. $\ln 5 + \ln 3$
2. $\ln 24 - \ln 8$
3. $\ln 5 - 1$

Example

Use the laws of logarithms to simplify :

1. $2\ln 7 - 3\ln 2$
2. $2\ln 3 + 3$

Example

Show that :

1. $\ln\left(\frac{1}{9}\right) = -2\ln 3$
2. $\ln 500 \approx 6.9078 - \ln 2$

EXERCISE 21

Write in the form $\ln a$, $a \in \mathbb{Q}$:

a $5\ln 3 + \ln 4$ **b** $3\ln 2 + 2\ln 5$ **c** $3\ln 2 - \ln 8$

d $3\ln 4 - 2\ln 2$ **e** $\frac{1}{3}\ln 8 + \ln 3$ **f** $\frac{1}{3}\ln(\frac{1}{27})$

g $-\ln 2$ **h** $-\ln(\frac{1}{2})$ **i** $-2\ln(\frac{1}{4})$

EXERCISE 22

Show that:

a $\ln \sqrt[3]{5} = \frac{1}{3}\ln 5$ **b** $\ln(\frac{1}{32}) = -5\ln 2$

c $\ln\left(\frac{1}{\sqrt[5]{2}}\right) = -\frac{1}{5}\ln 2$ **d** $\ln\left(\frac{e^2}{8}\right) = 2 - 3\ln 2$

Example

Write the following equations without logarithms :

1. $\ln A = 2\ln c + 3$
2. $\ln M = 3\ln a - 2$

EXERCISE 23

Write the following equations without logarithms, assuming all terms are positive:

a $\ln D = \ln x + 1$ **b** $\ln F = -\ln p + 2$ **c** $\ln P = \frac{1}{2}\ln x$

IV- Using logarithms

a) Exponential equations and logarithm

Example

Consider the equation $2^x = 30$.

(a) Solve for x, giving an exact answer, by using :

(i) base 2

(ii) base 10.

(b) Comment on your answers.

Example

Find x exactly :

(a) $e^x = 30$

(b) $3e^{x/2} = 21$

EXERCISE 24

Solve for x, giving an exact answer:

a $e^x = 10$ b $e^x = 1000$ c $2e^x = 0.3$

d $e^{\frac{x}{2}} = 5$ e $e^{2x} = 18$ f $e^{-\frac{x}{2}} = 1$

Example

Consider the equation $P = 200 \times 2^{0.04t}$.

(a) Rearrange the equation to give t in terms of P.

(b) Hence find the value of t when $P = 6$.

EXERCISE 25

Consider the equation $M = 20 \times 5^{-0.02x}$.

a Rearrange the equation to give x in terms of M.

b Hence find x when: i $M = 100$ ii $M = 232$

Example

Find algebraically the exact points of intersection of $y = e^x - 3$ and $y = 1 - 3e^{-x}$. Check your solution using technology.

EXERCISE 26

Solve for x:

a $e^{2x} = 2e^x$ b $e^x = e^{-x}$ c $e^{2x} - 5e^x + 6 = 0$

d $e^x + 2 = 3e^{-x}$ e $1 + 12e^{-x} = e^x$ f $e^x + e^{-x} = 3$

EXERCISE 27

Find algebraically the point(s) of intersection of:

a $y = e^x$ and $y = e^{2x} - 6$ b $y = 2e^x + 1$ and $y = 7 - e^x$

c $y = 3 - e^x$ and $y = 5e^{-x} - 3$

b) Change of base

Property

$\log_b a = \frac{\log_c a}{\log_c b}$ for $a,b,c > 0$ and $b,c \neq 1$.

Proof

If $\log_b a = x$, then $b^x = a$.

$\therefore \log_c(b^x) = \log_c a$ {taking logarithms in base c}

$\therefore x \log_c b = \log_c a$ {power law of logarithms}

$\therefore x = \dfrac{\log_c a}{\log_c b}$

$\therefore \log_b a = \dfrac{\log_c a}{\log_c b}$.

Example

Find $\log_2 9$ by :

1. letting $\log_2 9 = x$
2. changing to base 10
3. changing to base e.

Example

Solve for x exactly : $8^x - 5(4^x) = 0$.

EXERCISE 28

Use the rule $\log_b a = \dfrac{\log_{10} a}{\log_{10} b}$ to find, correct to 3 significant figures:

a $\log_3 12$ **b** $\log_{\frac{1}{2}} 1250$ **c** $\log_3(0.067)$ **d** $\log_{0.4}(0.006\,984)$

EXERCISE 29

Use the rule $\log_b a = \dfrac{\ln a}{\ln b}$ to solve, correct to 3 significant figures:

a $2^x = 0.051$ **b** $4^x = 213.8$ **c** $3^{2x+1} = 4.069$

EXERCISE 30

Solve for x exactly:

a $25^x - 3(5^x) = 0$ **b** $8(9^x) - 3^x = 0$ **c** $2^x - 2(4^x) = 0$

EXERCISE 31

Solve for x exactly: $\log_4 x^3 + \log_2 \sqrt{x} = 8$

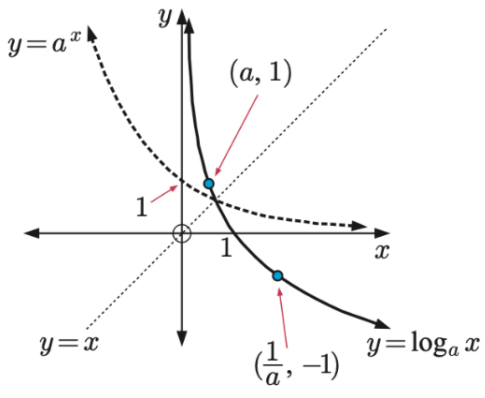
c) Graphs of logarithmic functions

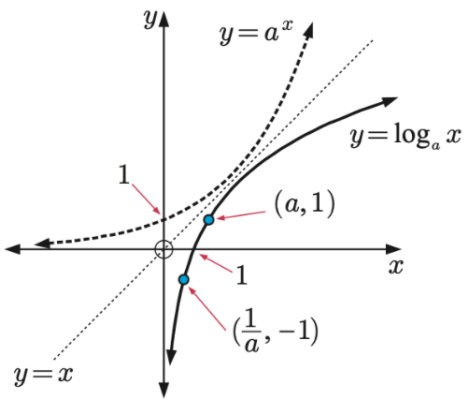
Definition

The inverse function f^{-1} is given by $x = a^y$, so $y = \log_a x$.

If $f(x) = a^x$ where $a > 0$, $a \neq 1$,then $f^{-1}(x) = \log_a x$.

Function	$f(x) = a^x$	$f^{-1}(x) = \log_a x$
Domain	$\{x \mid x \in \mathbb{R}\}$	$\{x \mid x > 0\}$
Range	$\{y \mid y > 0\}$	$\{y \mid y \in \mathbb{R}\}$
Asymptote	horizontal $y = 0$	vertical $x = 0$





The **vertical asymptote** of $y = \log_a x$ is the y-axis $x = 0$.

Property

In general, $y = \log_a\big(g(x)\big)$ is defined when $g(x) > 0$.

Example

Consider the function

$$f(x) = \log_2(x-1) + 1.$$

(a) Find the domain and range of f.

(b) Find any asymptotes and axes intercepts.

(c) Sketch the graph of f showing all important features.

(d) Find f^{-1} and explain how to verify your answer.

Example

Consider the function $f : x \mapsto e^{x-3}$.

(a) Find the equation defining f^{-1}.

(b) Sketch the graphs of f and f^{-1} on the same set of axes.

(c) State the domain and range of f and f^{-1}.

(d) Find any asymptotes and intercepts of f and f^{-1}.

EXERCISE 32

For the following functions f:

i Find the domain and range.
ii Find any asymptotes and axes intercepts.
iii Sketch the graph of $y = f(x)$ showing all important features.
iv Solve $f(x) = -1$ algebraically and check the solution on your graph.
v Find f^{-1} and explain how to verify your answer.

a $f : x \mapsto \log_3(x+1), \quad x > -1$
b $f : x \mapsto 1 - \log_3(x+1), \quad x > -1$
c $f : x \mapsto \log_5(x-2) - 2, \quad x > 0$
d $f : x \mapsto 1 - \log_5(x-2), \quad x > 2$

EXERCISE 33

Given $f : x \mapsto e^{2x}$ and $g : x \mapsto 2x - 1$, find:

a $(f^{-1} \circ g)(x)$
b $(g \circ f)^{-1}(x)$

EXERCISE 34

Consider the function $f : x \mapsto e^{x+3} + 2$.

a Find the defining equation for f^{-1}.
b Find the values of x for which:
i $f(x) < 2.1$ ii $f(x) < 2.01$ iii $f(x) < 2.001$ iv $f(x) < 2.0001$
Hence conjecture the horizontal asymptote for the graph of f.
c Determine the equation of the horizontal asymptote of $f(x)$ by discussing the behaviour of $f(x)$ as $x \to \pm\infty$.
d Hence, determine the vertical asymptote and the domain of f^{-1}.

d) Growth and decay

Population growth

Example

A farmer monitoring an insect plague notices that the area affected by the insects is given by

$$A_n = 1000 \times 2^{0.7n} \text{ hectares,}$$

where n is the number of weeks after the initial observation.

(a) Draw an accurate graph of A_n against n and use your graph to estimate the time taken for the affected area to reach 5000 ha.

(b) Check your answer to (a) using logarithms and using suitable technology.

Financial growth

Example

Iryna has €5000 to invest in an account that pays 5.2% p.a. interest compounded annually. Find, using logarithms, how long it will take for her investment to reach €20 000.

EXERCISE 35

The mass M_t of bacteria in a culture t hours after establishment is given by $M_t = 25 \times e^{0.1t}$ grams. Show that the time required for the mass of the culture to reach 50 grams is $10 \ln 2$ hours.

EXERCISE 36

Dien invests \$15 000 at 8.4% p.a. compounded *monthly*. He will withdraw his money when it reaches \$25 000, at which time he plans to travel. The formula $u_{n+1} = u_1 \times r^n$ can be used to model the investment, where n is the time in months.

a Explain why $r = 1.007$.

b After how many months will Dien withdraw the money?

EXERCISE 37

A man jumps from an aeroplane. His speed of descent is given by $V = 50(1 - e^{-0.2t})$ m s^{-1}, where t is the time in seconds. Show that it will take $5 \ln 5$ seconds for the man's speed to reach 40 m s^{-1}.

EXERCISE 38

The weight of radioactive substance remaining after t years is given by $W = 1000 \times 2^{-0.04t}$ grams.

a Sketch the graph of W against t.

b Write a function for t in terms of W.

c Hence find the time required for the weight to reach:

i 20 grams ii 0.001 grams.